



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 • STANDARD 6.AT.3

# Equivalent Ratios

Lesson 3-9 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can find and check equivalent ratios using multiplication and division.

**Language Objective:** I can explain my reasoning using the words equivalent ratios, rate, proportion, and simplify.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Equivalent Ratios

Ratio

Rate

Proportion

Simplify

Ratio table



Tap any word to see what it means and a picture.

1

Ratios that compare quantities in the same relationship are



2

A comparison of two quantities is a .

3

A ratio that compares two quantities with different units, like miles per hour, is a

4

An equation showing that two ratios are equal is a .

5

Writing a ratio with smaller numbers that compares the same way, like  $6:9 = 2:3$ , is to  it.

- 6 A table that lists equivalent ratios in rows or columns is a

## Write About the Math

A local smoothie shop sells a Berry Blast smoothie. The recipe calls for 2 cups of strawberries for every 3 cups of yogurt. Today they need to make 15 cups of yogurt worth of smoothies for a catering order?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Equivalent Ratios**  
*Razones equivalentes*

☐ **Ratio**  
*Razón*

☐ **Rate**  
*Tasa*

☐ **Proportion**  
*Proporción*

☐ **Simplify**  
*Simplificar*

**Model:** 32 is 4 times 8, so she multiplies both ingredients by 4. — A strong answer finds the scale factor 4 ( $32 \div 8$ ) and explains both ingredients must be multiplied by 4 to make equivalent ratios.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- Find the scale factor ( $15 \div 3$ ) and use it to figure out how many cups of strawberries the shop needs. Show your steps and explain why multiplying both ingredients by the same factor keeps the smoothie tasting the same.
- My answer is \_\_\_\_ because \_\_\_\_.

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.  
*Mi afirmación es \_\_\_\_.*
- My evidence is \_\_\_\_, so \_\_\_\_.  
*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

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### Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

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## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Equivalent Ratios
2. **Notes 2:** Ratio
3. **Notes 3:** Rate
4. **Notes 4:** Proportion
5. **Notes 5:** Simplify
6. **Notes 6:** Ratio table
7. **Watch for this mistake:** *A common mistake in Equivalent Ratios is ADDING the scale factor to one part instead of MULTIPLYING both parts by it. For example, scaling 2:5 by a factor of 4 should give 8:20 ( $2 \times 4$  and  $5 \times 4$ ) — not 6:9 ( $2+4$  and  $5+4$ ). Adding changes the taste of the recipe (and the value of the ratio); only multiplying or dividing both numbers by the same amount keeps it equivalent.*
8. **Try It 1:** A. 9:12 — *Multiply both parts of 3:4 by 3:  $3 \times 3 = 9$  and  $4 \times 3 = 12$ , so 9:12 is equivalent to 3:4.*
9. **Try It 2:** A. 15:6 — *Multiply both parts of 5:2 by 3:  $5 \times 3 = 15$  and  $2 \times 3 = 6$ , so 15:6 is equivalent to 5:2.*